

B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

February/March 2022 Semester End Main Examinations

Programme: B.E.

Branch: Computer Science And Engineering

Course Code: 20CS5PCSEG

Course: Software Engineering

Semester: V

Duration: 3 hrs.

Max Marks: 100

Date: 22.02.2022

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may suitably assumed.

UNIT - I

- 1
 - a) Define Software Engineering. Illustrate the key challenges faced by Software Engineering. 6
 - b) Giving reasons for your answer based on the type of system being developed, suggest the most appropriate generic software process model that might be used as a basis for managing the development of the following systems: 8
 - i) A system to control anti-lock braking in a car
 - ii) A virtual reality system to support software maintenance
 - iii) A university accounting system that replaces an existing system
 - iv) An interactive travel planning system that helps users plan journeys with the lowest environmental impact
 - c) State the ACM/IEEE 1999 Software Engineering code of ethics and professional practice 6

OR

- 2
 - a) Discuss the advantages and disadvantages of incremental development process. 6
 - b) List a set of non-functional requirements for the ticket-issuing system, setting out its expected reliability and response time and explain in detail with a neat diagram 8
 - c) Illustrate the Professional and ethical responsibilities that Software Engineer should have 6

UNIT - II

- 3
 - a) A software system is to be developed to manage the records of patients who enter a clinic for treatment. The records include records of all regular patient monitoring (temperature, blood pressure, etc.), treatments given, patient reactions and so on. After treatment, the records of their stay are sent to the patient's doctor who maintains their complete medical record. Identify the principal viewpoints which might be taken into account in the specification of this system and organize these using a viewpoint hierarchy diagram. 6

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.

- b) Using your knowledge of how an ATM is used, develop a set of use-cases, identify the interactions and explain the operations that could serve as a basis for understanding the requirements for an ATM system for i)Withdrawal of cash ii) Display balance. 8
- c) Develop a sequence diagram showing the interactions involved when a student registers for a course in a university. Courses may have limited enrolment, so the registration process must include checks that places are available. Assume that the student accesses an electronic course catalogue to find out about available courses. 6

UNIT - III

- 4 a) Describe Repository model with appropriate example. State the advantages and disadvantages of the repository model. 6
- b) Giving reasons for your answer, suggest an appropriate structural model for the following systems: 8
- i) An automated ticket-issuing system used by passengers at a railway station
 - ii) A computer-controlled video conferencing system that allows video, audio and computer data to be visible to several participants at the same time
 - iii) A robot floor-cleaner that is intended to clean relatively clear spaces such as corridors. The cleaner must be able to sense walls and other obstructions.
- c) Using the UML graphical notation for object classes, design State machine model for an automatic washing machine that models different programs for different types of clothes 6

UNIT - IV

- 5 a) An Organizations average productivity is 16FP/pm. The Average labor rate is \$5600 per month. if the proposed project has a count total of 479 and VAF is 56. Calculate 6
- i)Cost per Functional point
 - ii)Overall project cost
 - iii)Estimated effort in person months
- b) Define a Macroscopic scheduling. Draw time line for macroscopic project schedule for SafeHomeAssured.com webApp. 8
- c) Describe RMMM plan with respect to Risk Information sheet 6

UNIT - V

- 6 a) Illustrate with a neat diagram the tools that are included in RAD environment 6

b) A software manager is involved in the project development of a software design support system that supports the translation of software requirements to a formal software specification. Comment on the advantages and disadvantages of the following development strategies: 8

- i) Develop a throw-away prototype, evaluate it then review the system requirements. Develop the final system using C.
- ii) Develop the system from the existing requirements using Java, then modify it to adapt to any changed user requirements.
- iii) Develop the system using incremental development with a user involved in the development team.

c) Analyze the problems in developing performance tests for a distributed database system such as the LIBSYS system. 6

OR

7 a) Illustrate path testing approach of test case design by drawing a flow graph for a binary search routine. 6

b) Analyze why program inspections are an effective technique for discovering errors in a program. Write the types of error that are unlikely to be discovered through inspections. 8

c) Identify the strategic options for legacy system evolution. When would you normally replace all or part of a system rather than continue maintenance of the software (with or without reengineering) 6
